

EMMP-18-26

MC-54-19-21

M.Sc.: Physics

Course Title: Quantum Mechanics-I

Time Allowed 02:00 hours.

Shu 102

Course Code: PHYS-303

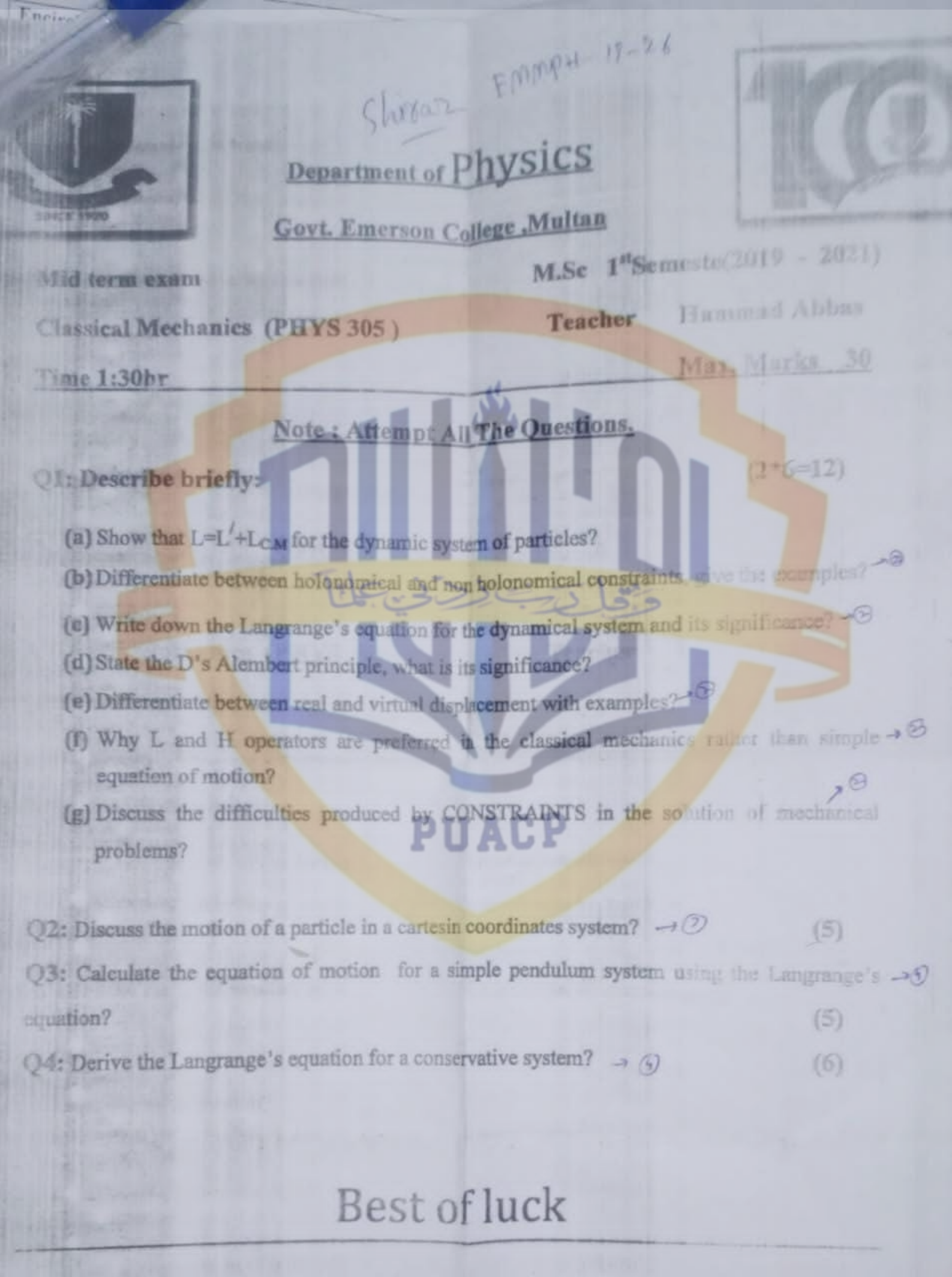
Session: 2019-2021 Semester 1st (Final)

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Max. Marks 60

No.	Questions	Marks
Q.1:	Encircle the correct answer.	(10)
1.	The commutation of position operator and momentum operator is given as (a) $i\hbar$ (b) $2i\hbar$ (c) $m\hbar$ (d) zero	
2.	The eigen value of z- component of angular momentum is (a) $m\hbar$ (b) $l(l+1)\hbar$ (c) $(m+1)\hbar$ (d) $(m+1)\hbar$	
3.	Due to raising operator, the eigen value of harmonic oscillator is increase to (a) $i\hbar\omega$ (b) $m\hbar\omega$ (c) $\hbar\omega$ (d) remain same	
4.	If two vectors are normalized and orthogonal also we called them _____ vectors. (a) Normalized (b) Orthogonal (c) Orthonormal (d) All of above	
5.	The bra space is _____ to ket space: (a) Hilbert space (b) Vector space (c) Dual space (d) None of these	
6.	The eigen value of Hermitian operator is always (a) real (b) imaginary (c) orthogonal (d) complex	
7.	The concept of spin motion of electron in an atom is based on (a) classical model (b) quantum motion (c) planetary motion (d) all of these	
8.	Expression for z-component of angular momentum is: (a) $-i\hbar \frac{\partial}{\partial \phi}$ (b) $-i\hbar \frac{\partial}{\partial \theta}$ (c) $-i\hbar \frac{\partial}{\partial z}$ (d) All of these	
9.	It is necessary for two operators to commute with each other for having (a) Degenerate eigen values (b) Simultaneous eigen function (c) Orthogonal eigen (d) All of these	
10.	Radial equation for central potential depends on _____ quantum number (a) Azimuthal (b) Magnetic (c) Spin (d) None	
Q.2:	Write short answer the following questions.	(30)
1.	How we differentiate Hilbert space from linear vector space.	
2.	Find the Hermitian adjoint of function $f(A) = (1+iA+3A^2)(1-2iA-9A^2)/(5+7A)$.	
3.	Show that the operator $ \Psi\rangle\langle\Psi $ is projection operator only when $ \Psi\rangle$ is normalized.	
4.	Write the Eigen value and represent them in matrices for raising and lowering operator for angular momentum.	
5.	Write classical, quantum and experimental point of view about spin of electron and also write pauli spin matrices for spin angular momentum.	
6.	What do you know about space translation and time translation and write their generator and results.	
7.	Describe the radial equation for hydrogen atom and explain physical significance of radial equation.	
8.	Write the generalized form for Hsienberg uncertainty principle and discuss results for special case.	

P.T.O.



Shirbaz
F.M.P.H. 17-26

Department of **Physics**

Govt. Emerson College, Multan

Mid term exam

M.Sc 1st Semester (2019 - 2021)

Classical Mechanics (PHYS 305)

Teacher

Hammad Abbas

Time 1:30hr

Max. Marks 30

Note: Attempt All The Questions.

Q1: Describe briefly:

(2*6=12)

- (a) Show that $L=L' + L_{CM}$ for the dynamic system of particles?
- (b) Differentiate between holonomical and non holonomical constraints. Give the examples?
- (c) Write down the Lagrange's equation for the dynamical system and its significance?
- (d) State the D's Alembert principle, what is its significance?
- (e) Differentiate between real and virtual displacement with examples?
- (f) Why L and H operators are preferred in the classical mechanics rather than simple equation of motion?
- (g) Discuss the difficulties produced by CONSTRAINTS in the solution of mechanical problems?

Q2: Discuss the motion of a particle in a cartesian coordinates system? → ②

(5)

Q3: Calculate the equation of motion for a simple pendulum system using the Lagrange's equation? → ③

(5)

Q4: Derive the Lagrange's equation for a conservative system? → ④

(6)

Best of luck

FMMPH - 18-26

Shiraz

MC-53-19-21

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Bahauddin Zakariya University Multan

Program: M.Sc. (Physics)Semester: 1st (Final Term)Session: 2019-2021Course Title: Methods of Mathematical Physics-ICourse Code: PHYS-301Time Allowed: 02:30 hrsMax. Marks: 60

Note: Attempt all the Questions:

Q.No.	Questions	Marks
Q.1:	Write short answer of the following.	
	i). Define Cyclic - group. ii). Show that Kronecker Delta δ_k^i is really a mixed tensor of rank =2 ? iii). Stoke's Theorem iv). State Cauchy's Residue Theorem.	4x3=12
Q.2:	a) Verify Green theorem for $\oint_C (xy + y^2) dx + x^2 dy$ for closed path enclosed by closed curves $y = x$ and $y = x^2$. b) Find $\nabla \cdot [f(r) \mathbf{r}^k]$, where $f(r) = r^n$ and $\mathbf{r}^k = \frac{x}{r} \mathbf{e}_1 + \frac{y}{r} \mathbf{e}_2 + \frac{z}{r} \mathbf{e}_3$?	8 8
Q.3:	a) Derive unit vectors in spherical coordinate system and prove that these are mutually orthogonal. b) If R is a matrix formed by eigen vectors $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, \mathbf{r}_n)$ whose corresponding eigen value are $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \dots, \lambda_n$ then prove that $R^{-1}AR$ is a diagonal matrix.	8 8
Q.4:	a) Is $f(z) = z^2 - 3z + 2$ analytical? Using calculus of residues prove that $\int_0^{2\pi} \frac{d\theta}{1 - 2t \cos \theta + t^2} = \frac{2\pi}{1 - t^2} \quad -1 < t < 1$	8 8

PUACP

MC-53-19-21-200.

Govt. Emerson College Multan

Program: M.Sc (Physics)

Course: Methods of Mathematical Physics-I

Time allowed: 1:30 hours

Semester: I (Mid. Term)

Marks: 30

Course Code: PHYS-301

Course Instructor: Ms. Sabia Iqbal

PUACP

Attempt all questions

(4 x 3 = 12)

Q.NO:1 Describe briefly.

- What is physical interpretation of Gradient of a scalar?
- Prove that curl of tangential velocity of a rotating body is twice that of angular velocity.
- Euler's formula and De Moivre's formula.
- Significance of complex numbers.

Q.NO:2 Derive Cauchy Riemann conditions in polar form.

Q.NO:3 Define Residue...Also state and prove Cauchy's Residue Theorem.

Q.No.4 Verify Green's theorem for closed region $\int (xy+y^2)dx + x^2 dy$ which is bounded by $y=x$ and $y=x^2$

Note: All symbols used have their usual meaning.

Department of physics

Mid Term Exam 2020

Govt Emerson College Multan

Course Title: Analog Electronics I

Course Code: PHYS-309

Session: 2019-2021

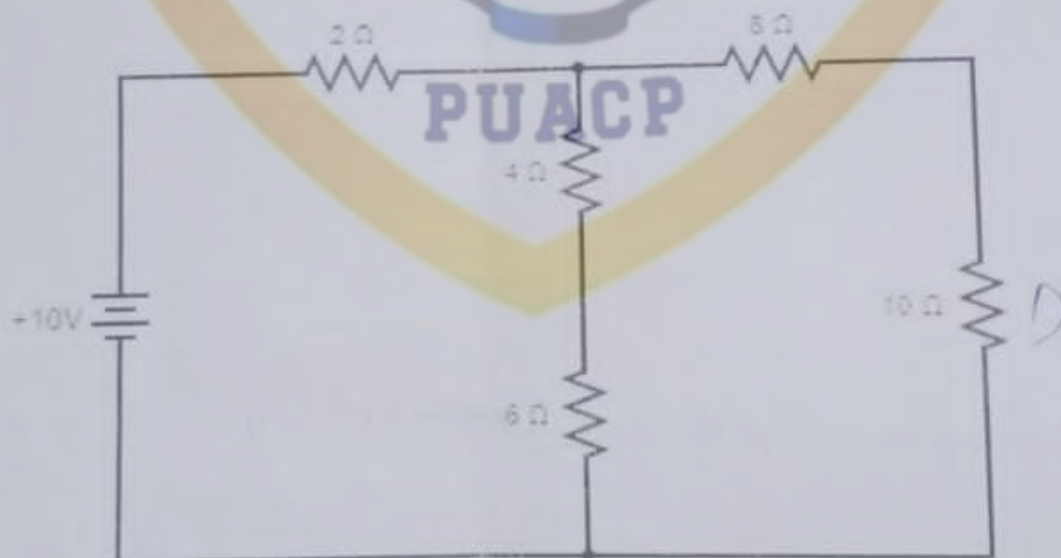
Time: 1 hr. 30 min Total Marks: 30

Short Questions: 2x8=16

- ✓1. Two equal value resistors are connected in a series across a 15V battery. What is the voltage drop across second resistor?
- ✓2. If you want to obtain a total resistance of $130\ \Omega$, what can you add in parallel to parallel combination of $330\ \Omega$ & $270\ \Omega$?
- ✓3. Develop a circuit consisting of a voltage source, reverse biased diode & limiting resistor.
4. To more accurately represent a diode what factors must be included?
- ✓5. Why is the superposition theorem useful for analysis of multiple source circuits?
- ✓6. What are the two basic components of a Thevenin's circuit?
- ✓7. If a temperature of a silicon diode is changed by 10°C from room temperature, what will be barrier potential of silicon diode?
- ✓8. Find the average value of a full wave rectified voltage if its output peak is 10V?

Long Questions:

- ✓2. What is linear network? Give one example of a linear network by using circuit diagram and explain it briefly. (4)
- ✓3. What is diode? Draw its block diagram and its schematic symbol. Discuss the various factors that can affect the barrier potential of a diode. (4)
- ✓4. Determine the total resistance of a network shown in figure. (6)



GOVT. EMERSON COLLEGE MULTAN

EMMPH-19-26

Slu802

Subject: Computational Physics

Course Code: PHYS-307

Total Marks: 30

Mid-term Exam M.Sc. (Physics) 1st Semester (2019-2021)

Total Time: 120 Hrs.

Q1 Define Short Questions (Describe the questions briefly)

- I. Define desk checking?
- II. List some advantages and disadvantages of machine language? (2)
- III. Which language is used to write algorithms? (2)
- IV. Write the difference between structured and object oriented programming? Also clarify (2) that C++ belongs to which one of the category?
- V. What is a syntax error? (2)
- VI. Why is logical error the most difficult error? (2)
- VII. What is keyword also give an example? (2)
- VIII. What is the difference between the preincrement operators ++x and post increment operator x++? (2)
- IX. Define operator precedence with example?
- X. Write a program that can display your name on the output? (2)

Long Questions

- Q2** Describe some debugging features of the turbo C++? (5)
- Q3** Write a program to calculate and print the area of square with given height and width (5)

Watches, modify, breakpoint.
Logical, Sytan, Runtime.